

OWASE, Japan

WMO: 476630

Lat: 34.0697N

Lon: 136.1933E

Elev: 17

StdP: 101.12

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	0.5	-10.7	1.5	5.0	-9.1	1.7	5.7	9.5	7.6	8.6	7.6	1.9	260	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	33.0	25.4	31.6	25.3	30.4	25.0	26.6	30.5	26.1	29.8	25.7	29.2	4.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	20.8	28.5	25.1	20.2	28.1	24.6	19.7	27.7	83.3	30.6	81.2	30.0	79.2	29.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.5	5.7	-2.0	36.0	0.8	1.1	-2.5	36.8	-3.0	37.5	-3.4	38.1	-4.0	39.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-4.0	27.4	0.8	0.4	-4.6	27.7	-5.1	27.9	-5.6	28.1	-6.2	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.6	7.3	10.2	14.7	18.8	22.0	26.1	27.1	24.1	19.2	13.9	9.0
	DBStd	7.45	2.19	2.71	2.91	2.79	2.02	2.36	2.12	1.78	2.25	2.45	2.85	2.76
	HDD10.0	284	110	84	33	2	0	0	0	0	0	2	53	
	HDD18.3	1501	365	309	252	113	19	1	0	0	19	134	289	
	CDD10.0	2702	3	8	41	143	272	359	498	531	423	285	119	21
	CDD18.3	876	0	0	0	3	32	109	240	272	173	45	2	0
	CDH23.3	6650	0	0	1	11	117	499	2083	2686	1106	145	2	0
	CDH26.7	1954	0	0	0	1	11	108	647	931	247	11	0	0
Wind	WSAvg	2.3	2.5	2.4	2.5	2.4	2.3	2.0	2.1	2.3	2.2	1.9	2.0	2.4
Precipitation	PrecAvg	3355	97	116	209	252	328	370	321	365	630	348	223	98
	PrecMax	5083	356	288	406	533	828	759	704	906	1147	848	911	262
	PrecMin	1501	1	10	67	62	118	148	43	7	128	50	26	4
	PrecStd	745	82	74	99	123	189	128	172	244	278	199	209	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.4	21.6	24.7	28.0	32.8	34.8	35.2	32.8	28.2	23.0	19.3
		MCWB	10.7	12.6	13.6	17.1	18.9	24.2	25.3	25.7	24.6	21.1	17.2	14.1
	2%	DB	14.0	15.8	18.9	22.6	25.9	29.5	32.7	33.3	30.5	26.3	21.5	17.0
		MCWB	9.3	11.4	12.8	15.9	18.7	23.1	25.4	25.7	24.0	20.4	16.3	12.3
	5%	DB	12.7	14.0	17.3	21.2	24.5	27.4	31.2	31.9	29.1	24.9	20.3	15.5
		MCWB	8.1	9.5	12.1	15.5	18.1	22.0	25.2	25.6	23.7	19.6	15.6	10.9
	10%	DB	11.4	12.6	15.8	20.0	23.3	26.0	29.9	30.8	28.0	23.7	19.0	14.1
		MCWB	6.8	8.3	11.1	14.9	17.8	21.5	24.9	25.4	23.2	18.8	14.9	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	14.9	16.4	20.3	22.6	25.1	27.0	27.3	26.3	23.9	19.7	16.4
		MCDB	14.2	16.6	18.2	22.5	25.2	29.4	31.1	31.3	29.5	25.9	21.1	17.8
	2%	WB	10.7	12.7	14.8	18.4	21.0	24.2	26.3	26.7	25.6	22.5	18.3	13.8
		MCDB	12.4	14.5	17.2	20.6	23.6	27.9	30.3	30.6	28.4	24.4	19.9	15.8
	5%	WB	9.1	10.9	13.5	17.1	20.0	23.5	25.8	26.4	25.1	21.4	17.2	12.2
		MCDB	11.4	12.9	15.9	19.7	22.6	26.6	29.6	30.2	27.6	23.3	18.9	14.1
	10%	WB	7.7	9.3	12.3	16.0	19.1	22.6	25.4	26.0	24.5	20.3	16.1	10.8
		MCDB	10.5	11.5	14.8	18.8	21.7	25.1	28.9	29.6	26.9	22.3	18.0	13.1
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.1	9.3	8.9	8.0	6.3	6.2	6.6	6.7	7.7	8.8	8.9
		MCDBR	10.2	10.4	10.7	10.5	10.4	8.7	8.5	8.6	8.1	9.0	9.5	9.8
		MCWBR	6.8	7.3	7.1	6.3	5.7	3.9	3.4	3.7	4.1	4.9	6.0	6.5
	5% WB	MCDBR	8.6	8.9	8.9	8.2	7.8	7.1	7.3	7.2	6.4	6.7	7.3	8.2
	MCWBR	6.9	7.2	7.1	5.7	5.1	3.4	3.4	3.6	3.5	4.5	5.7	6.8	
Clear-Sky Solar Irradiance	taub	0.391	0.438	0.500	0.540	0.549	0.559	0.558	0.525	0.496	0.467	0.419	0.387	
	taud	2.200	2.083	1.932	1.886	1.935	1.987	2.035	2.117	2.167	2.164	2.203	2.226	
	Ebn at Noon	793	792	773	764	762	751	749	768	771	756	756	771	
	Edh at Noon	114	143	180	197	190	180	171	155	141	131	114	106	
All-Sky Solar Radiation	RadAvg	2.58	3.22	4.17	4.87	5.19	4.55	5.18	5.31	4.14	3.24	2.65	2.36	
	RadStd	0.18	0.25	0.29	0.51	0.58	0.46	0.63	0.43	0.35	0.35	0.26	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page